

# Experiment 8

## TCP vs UDP Performance & Reliability

### Lab 8 Objectives:

- Establish connections using TCP and UDP, compare reliability and performance.

**Prerequisites:** The prerequisites of this lab are-

- Basic understanding of TCP and UDP protocols.
- Familiarity with network sockets and data transmission.

**Outcomes:** By the end of this lab, students will be able to:

- Establish and test connections using TCP and UDP.
- Compare data reliability, latency, and throughput.
- Analyze packet loss and retransmission behavior.

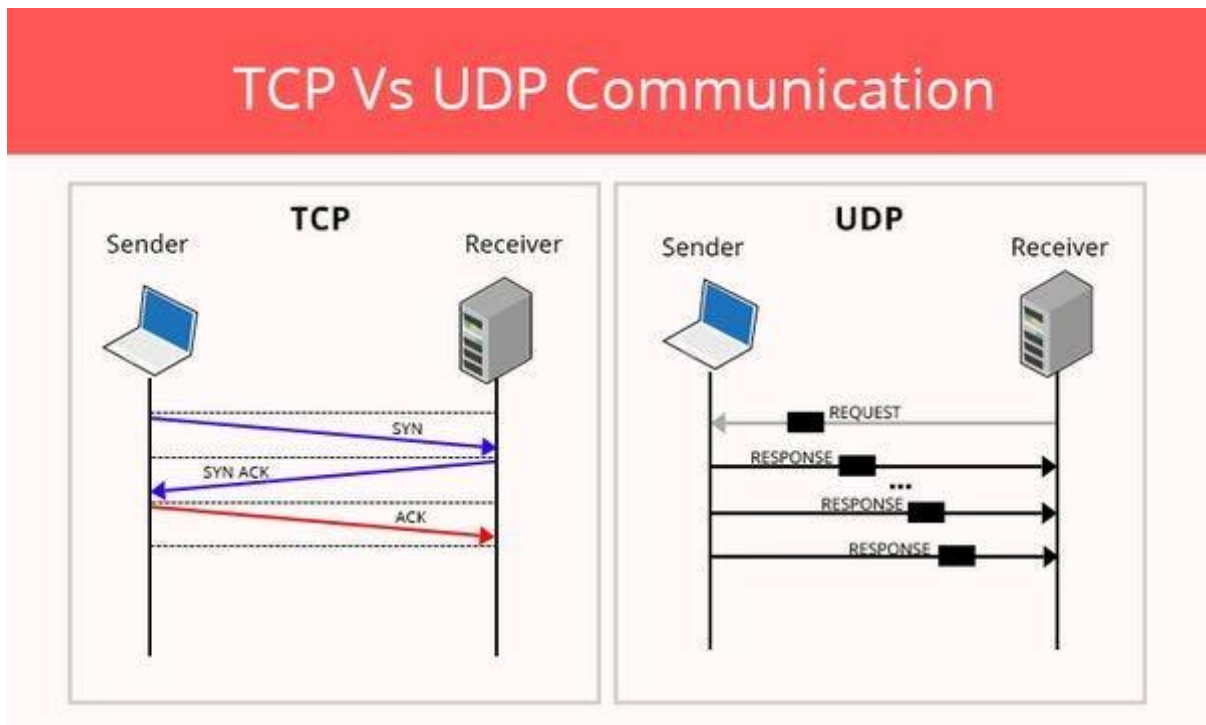
### TCP vs UDP Protocols

| Feature           | TCP                                                           | UDP                                     |
|-------------------|---------------------------------------------------------------|-----------------------------------------|
| Connection Type   | Connection-oriented                                           | Connectionless                          |
| Reliability       | Reliable data transfer with acknowledgment and retransmission | Unreliable, no guarantee of delivery    |
| Ordering          | Data packets arrive in order                                  | No ordering guarantees                  |
| Error Handling    | Error detection, retransmission, flow control                 | Basic error checking, no retransmission |
| Speed             | Slower due to overhead                                        | Faster, less overhead                   |
| Use Case Examples | Web browsing, email, file transfer                            | Video streaming, online games           |

## Procedure

### Step 1: Setup Client-Server Model for TCP and UDP

- Create a **TCP server** and **TCP client** that establish a connection and exchange data.
- Create a **UDP server** and **UDP client** to send and receive the same data without connection establishment.



### Step 2: Establish TCP Connection and Transfer Data

- TCP client connects to the server.
- Data is sent, and server acknowledges receipt.
- Ensure retransmission happens if packets are lost.
- Measure latency and throughput during data transfer.

### Step 3: Establish UDP Connection and Send Data

- UDP client sends data packets to the server.

- No connection or acknowledgment.
- Measure latency and throughput for UDP transfer.

#### Step 4: Measure and Compare Performance Metrics

| Metric      | TCP                                               | UDP                         |
|-------------|---------------------------------------------------|-----------------------------|
| Latency     | Higher due to connection setup and error checking | Lower, connectionless       |
| Throughput  | Moderate, due to retransmissions and flow control | Higher, less overhead       |
| Packet Loss | Low, retransmissions recover lost packets         | Higher, no retransmission   |
| Reliability | High, ensures data delivery                       | Low, no delivery guarantees |
| Ordering    | Guaranteed                                        | Not guaranteed              |

#### Step 5: Analyze Packet Loss and Retransmission Behavior

- Use tools like **Wireshark** or **Ncat** to capture packets.
- Observe retransmission packets in TCP if any loss occurs.
- UDP packets lost are not retransmitted.
- Analyze how retransmissions affect overall performance and latency in TCP.

#### Sample Commands and Code Snippets

##### Netcat TCP Server

```
nc -l 1234
```

##### Netcat TCP Client

```
nc 192.168.1.10 1234
```

## Netcat UDP Server

```
nc -u -l 1234
```

## Netcat UDP Client

```
nc -u 192.168.1.10 1234
```

## Observations and Results

- TCP ensures reliable data transfer but has more overhead, resulting in higher latency.
- UDP is faster but does not guarantee data delivery.
- Packet loss is observed in UDP, but TCP handles loss through retransmission.
- Depending on the application, either TCP or UDP is chosen balancing reliability and speed.

## TCP in Wireshark

| tcp                                                                                                                                                         |                |                |          |        |        |                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------|----------|--------|--------|-----------------------------------------------------------------------------------------|
| Time                                                                                                                                                        | Source         | Destination    | Protocol | Length | No.    | Info                                                                                    |
| 3450.5444...                                                                                                                                                | 204.79.197.222 | 192.168.226.42 | TLSv1.3  | 85     | 185620 | Application Data                                                                        |
| 3450.5444...                                                                                                                                                | 192.168.226.42 | 204.79.197.222 | TCP      | 54     | 185621 | 57929 → 443 [ACK] Seq=1976 Ack=7131 Win=261120 Len=0                                    |
| 3450.5997...                                                                                                                                                | 204.79.197.222 | 192.168.226.42 | TCP      | 54     | 185622 | 443 → 57929 [ACK] Seq=7131 Ack=1976 Win=4194560 Len=0                                   |
| 3450.6710...                                                                                                                                                | 13.69.116.107  | 192.168.226.42 | TCP      | 66     | 185623 | 443 → 57930 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1275 WS=256 SACK_PERM            |
| 3450.6711...                                                                                                                                                | 192.168.226.42 | 13.69.116.107  | TCP      | 54     | 185624 | 57930 → 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0                                          |
| 3450.6789...                                                                                                                                                | 192.168.226.42 | 13.69.116.107  | TLSv1.3  | 611    | 185625 | Client Hello (SNI=self.events.data.microsoft.com)                                       |
| 3450.9515...                                                                                                                                                | 13.69.116.107  | 192.168.226.42 | TLSv1.3  | 1329   | 185626 | Server Hello, Change Cipher Spec                                                        |
| 3450.9515...                                                                                                                                                | 13.69.116.107  | 192.168.226.42 | TCP      | 1329   | 185627 | 443 → 57930 [ACK] Seq=1276 Ack=558 Win=4194048 Len=1275 [TCP PDU reassembled in 185631] |
| 3450.9515...                                                                                                                                                | 13.69.116.107  | 192.168.226.42 | TCP      | 1329   | 185628 | 443 → 57930 [ACK] Seq=2551 Ack=558 Win=4194048 Len=1275 [TCP PDU reassembled in 185631] |
| 3450.9515...                                                                                                                                                | 13.69.116.107  | 192.168.226.42 | TCP      | 1329   | 185629 | 443 → 57930 [ACK] Seq=3826 Ack=558 Win=4194048 Len=1275 [TCP PDU reassembled in 185631] |
| 3450.9515...                                                                                                                                                | 13.69.116.107  | 192.168.226.42 | TCP      | 1329   | 185630 | 443 → 57930 [ACK] Seq=5101 Ack=558 Win=4194048 Len=1275 [TCP PDU reassembled in 185631] |
| 3450.9515...                                                                                                                                                | 13.69.116.107  | 192.168.226.42 | TLSv1.3  | 84     | 185631 | Application Data                                                                        |
| 3450.9516...                                                                                                                                                | 192.168.226.42 | 13.69.116.107  | TCP      | 54     | 185632 | 57930 → 443 [ACK] Seq=558 Ack=6406 Win=262144 Len=0                                     |
| 3450.9619...                                                                                                                                                | 192.168.226.42 | 13.69.116.107  | TLSv1.3  | 134    | 185633 | Change Cipher Spec, Application Data                                                    |
| 3450.9621...                                                                                                                                                | 192.168.226.42 | 13.69.116.107  | TLSv1.3  | 505    | 185634 | Application Data                                                                        |
| 3450.9622...                                                                                                                                                | 192.168.226.42 | 13.69.116.107  | TLSv1.3  | 2791   | 185635 | Application Data                                                                        |
| 3451.2599...                                                                                                                                                | 13.69.116.107  | 192.168.226.42 | TLSv1.3  | 157    | 185636 | Application Data                                                                        |
| 3451.2599...                                                                                                                                                | 13.69.116.107  | 192.168.226.42 | TLSv1.3  | 500    | 185637 | Application Data                                                                        |
| 3451.2599...                                                                                                                                                | 13.69.116.107  | 192.168.226.42 | TCP      | 54     | 185638 | 443 → 57930 [ACK] Seq=6509 Ack=3639 Win=4194560 Len=0                                   |
| 3451.2599...                                                                                                                                                | 13.69.116.107  | 192.168.226.42 | TCP      | 54     | 185639 | 443 → 57930 [ACK] Seq=6509 Ack=3826 Win=4194560 Len=0                                   |
| 3451.2600...                                                                                                                                                | 192.168.226.42 | 13.69.116.107  | TCP      | 54     | 185640 | 57930 → 443 [ACK] Seq=3826 Ack=6955 Win=261376 Len=0                                    |
| 3451.2600...                                                                                                                                                | 192.168.226.42 | 13.69.116.107  | TCP      | 54     | 185641 | [TCP Dup ACK 185640#1] 57930 → 443 [ACK] Seq=3826 Ack=6955 Win=261376 Len=0             |
| ▶ Frame 1005: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{7EC3D058-0010-0030-8000-000000000000} (0.0000000 seconds) |                |                |          |        |        |                                                                                         |
| ▶ Ethernet II, Src: Intel_00:10:13 (dc:45:46:a0:10:13), Dst: 4e:a9:95:a0:5c:c0 (4e:a9:95:a0:5c:c0)                                                          |                |                |          |        |        |                                                                                         |
| ▶ Internet Protocol Version 4, Src: 192.168.226.42, Dst: 23.208.176.183                                                                                     |                |                |          |        |        |                                                                                         |
| ▶ Transmission Control Protocol, Src Port: 56924, Dst Port: 443, Seq: 2614, Ack: 53249, Len: 0                                                              |                |                |          |        |        |                                                                                         |
| 0000 4e a9 95 a0 5c c0 dc 45 46 a0 10 13 08 00 45 00 N \ E F ... E                                                                                          |                |                |          |        |        |                                                                                         |
| 0010 00 34 79 ab 40 00 80 06 00 00 c0 a8 e2 2a 17 d0 4y @ ... *                                                                                             |                |                |          |        |        |                                                                                         |
| 0020 b0 b7 de 5c 01 bb b2 b0 5c d4 81 b6 c8 94 80 10 ... \ ... \ ...                                                                                        |                |                |          |        |        |                                                                                         |
| 0030 00 ff 6b 81 00 00 01 01 05 0a 81 b6 c4 36 81 b6 k ... 6...                                                                                             |                |                |          |        |        |                                                                                         |
| 0040 c8 94                                                                                                                                                  |                |                |          |        |        |                                                                                         |